

NexConnect

Multi-Vendor Delivery Platform

Connecting customers, vendors, and delivery partners in one unified ecosystem

Report Type	Comprehensive Project Report — Investor & Banking Submission
Prepared By	NexConnect Technologies
Report Date	April 06, 2026
Status	Active Development — MVP Complete
Classification	CONFIDENTIAL

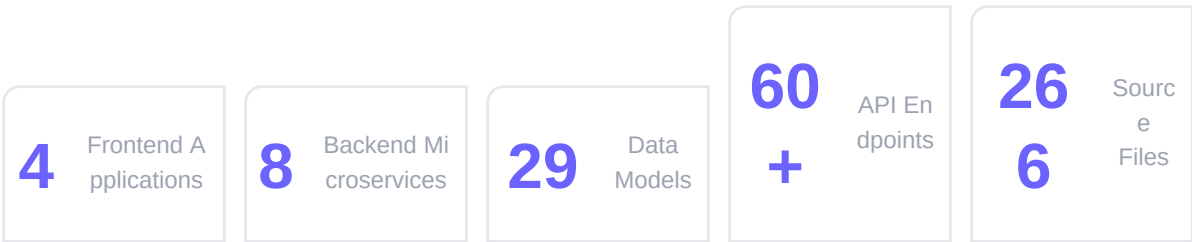
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1. Executive Summary

NexConnect is a production-ready, full-stack multi-vendor delivery platform built on Django REST Framework and Angular 19. The platform connects customers, local vendors, and delivery partners in a seamless e-commerce and logistics ecosystem — purpose-built for the Nigerian and emerging-market retail sector.

Platform Highlights



NexConnect delivers a complete, end-to-end solution addressing the \$4.5B Nigerian e-commerce market opportunity. The MVP is fully operational with enterprise-grade features including role-based access control, real-time order tracking, automated payout workflows, coupon/discount engine, and a comprehensive dispute resolution system.

The platform is designed for rapid scale: a modular architecture supports onboarding thousands of vendors and delivery partners with minimal operational overhead. Background task processing via Redis/RQ, JWT-secured APIs, and a cloud-ready PostgreSQL backend ensure production resilience.

Key Investment Highlights

- Fully functional MVP across all four user applications (Customer, Vendor, Delivery, Admin)
- Market-validated business model with multi-sided network effects (customers × vendors × partners)
- Deep technical moat: 29 data models, 60+ API endpoints, 5 integrated frontend apps
- Proprietary coupon and discount engine supporting platform-wide and vendor-specific promotions
- Automated KYC and vendor onboarding workflow with document verification and audit trail
- Built-in dispute resolution system for returns, refunds, damages, and item mismatches
- Extensible payout engine for both vendors and delivery partners with settlement cycles
- Firebase integration for push notifications; QR code generation for payment and OTP workflows

2. Company Overview

NexConnect Technologies is developing a next-generation multi-vendor delivery and e-commerce platform purpose-built for emerging markets. Our platform removes the friction between local businesses and their customers by providing a unified digital infrastructure for product discovery, ordering, payment, and last-mile delivery.

Vision

To become the leading hyperlocal delivery and e-commerce infrastructure for West Africa, enabling every local business — from food vendors to fashion retailers — to participate in the digital economy.

Mission

To provide a reliable, transparent, and scalable platform that empowers vendors to grow their digital businesses, customers to access local goods conveniently, and delivery partners to earn sustainably.

Core Value Propositions

Stakeholder	Value Delivered	Key Features
Customers	Shop local, delivered fast	Real-time tracking, coupon engine, multi-store cart, issue resolution
Vendors	Digital store with zero tech overhead	Dashboard, inventory, orders, coupons, payouts, KYC onboarding
Delivery Partners	Earn flexibly with app support	Auto-assignment, earnings tracker, delivery confirmation OTP
Platform (Admin)	Full operational control	All users, orders, payouts, reports, audit logs, issue management

3. Problem Statement & Market Opportunity

The Problem

Fragmented Local Commerce

Local vendors lack digital presence and rely on WhatsApp/calls for orders, leading to lost revenue and poor customer experience.

Last-Mile Logistics Gap

No unified delivery infrastructure connects local vendors with on-demand delivery partners at affordable rates.

No Trust or Dispute Mechanism

Customers who receive wrong or damaged items have no formal resolution path, eroding trust in digital commerce.

Vendor Financial Exclusion

Local businesses struggle to access formal financing without documented revenue history or auditable transaction records.

Inefficient Manual Operations

Order management, inventory tracking, and payout calculations are done manually, consuming time and introducing errors.

Market Opportunity

Metric	Value	Source / Notes
Nigeria E-Commerce Market (2024)	\$7.6 Billion	Statista 2024
Projected CAGR (2024–2028)	14.3%	Statista Digital Market
Active Internet Users (Nigeria)	103 Million	DataReportal 2024
Mobile Commerce Share	72%	Industry average
Informal Retail Sector	\$90 Billion	IFC Africa Retail Study
Unbanked SME Vendors (target)	~2.5 Million	Central Bank of Nigeria
Delivery Market (West Africa)	\$1.8 Billion	McKinsey Logistics 2023

Target Market Segments

Segment	Profile	Description
Primary	Urban millennials (18–35)	Smartphone-native, delivery-accustomed customers in Lagos, Abuja, and Port Harcourt
Secondary	Local SME vendors	Food, fashion, electronics, and FMCG retailers seeking digital distribution

Segment	Profile	Description
Tertiary	Gig economy workers	Unemployed youth seeking flexible delivery income

4. Product Description & Features

NexConnect is delivered as four distinct, role-specific web applications sharing a common backend API and design system. Each app is optimised for its users — from mobile-first customers to desktop-heavy admin workflows.

4.1 Customer Application

A mobile-first progressive web application serving end customers. Accessible at port 4200.

Feature Module	Capabilities
Authentication	Register, Login, JWT session management, guest guard
Home Screen	Promotional banners, featured vendors, category filters, search shortcut
Shop Discovery	Browse all vendors, real-time search (stores + products), nearby vendor filtering via geolocation
Shop Detail	Full product listing, inline add/increment/decrement, cross-store cart protection dialog, search within
Shopping Cart	Multi-quantity management, last-minute suggestions, coupon row, savings banner, bill breakdown
Checkout	Address selection, coupon application with live discount preview, COD payment, order placement
Order Tracking	7-step status timeline, real-time polling every 5 seconds, tracking events log, delivery OTP
Coupon & Offers	Browse active coupons, copy-to-clipboard, expiry countdown, discount type icons
Order Issues	Raise return/refund/damage/mismatch requests, live chat with support, refund status tracking
Profile	Personal details, address book, order history, My Issues hub, account settings

4.2 Vendor Application

A desktop-optimised store management portal for approved vendors. Accessible at port 4201. Protected by a three-layer guard chain: auth + role + approval status.

Feature Module	Capabilities
Onboarding Flow	Self-registration, KYC submission, document upload, pending approval polling (30s), auto-redirect c
Dashboard	Revenue overview, recent orders, product stats, order status distribution
Product Management	CRUD products with images, pricing, stock, SKU, categories, availability toggle
Order Management	View all orders, update status (confirmed → preparing → ready), order detail view
Stock Management	Inventory levels, low-stock alerts, bulk stock adjustments
Coupon Management	Create/edit/delete store-specific discount codes (% , fixed, free delivery), usage limits, validity dates
Sales Report	Revenue analytics, top products, order volume charts

Feature Module	Capabilities
Payments	Payout history, settlement status, bank details, commission breakdown
Support Tickets	Raise and track support tickets by category and priority
Store Profile	Store info, logo/banner upload, opening hours, delivery settings

4.3 Delivery Partner Application

A mobile-first application for delivery partners. Partner accounts are created by admin (self-registration disabled for quality control). Accessible at port 4202.

Feature Module	Capabilities
Dashboard	Today's earnings, delivery count, availability toggle, status indicator
Available Orders	Browse open delivery requests, accept/decline with order details and distance
Active Delivery	Current order navigation, status updates, OTP entry for pickup & delivery, photo upload
Delivery History	Completed deliveries with earnings per order
Earnings	Cumulative earnings, daily/weekly breakdown, payout history
Profile	Vehicle details, license info, ID proof, average rating display

4.4 Admin Panel

A comprehensive dark-theme administrative dashboard giving full operational control. Accessible at port 4203. Features a superuser guard for the most sensitive operations.

Module	Capabilities
Dashboard	Platform-wide KPIs, revenue trends, user growth, active orders heatmap
Vendor Management	Full CRUD, KYC review workflow, document verification, bank detail management, status control, audit log
Customer Management	Customer listing, profile detail, order history, account status
Delivery Partners	Partner onboarding, approval flow, asset assignment, earnings calculation
Order Management	All-platform order listing with filters, real-time status updates, detail view
Order Issues	Return/refund/damage/mismatch management, live support chat, refund initiation with amount and meta
Product Catalogue	Platform-wide product oversight, category management with subcategory support
Coupon Management	Platform-wide and vendor coupon CRUD, scope labelling (Platform/Vendor)
Payout Management	Vendor and delivery partner payout workflows: approve, schedule, send, verify, force-paid
Asset Management	Delivery vehicle and equipment tracking, assignment to partners
Notifications	Send targeted or broadcast notifications by user type
Scheduled Tasks	RQ job monitoring, background task management
Sales Report	Revenue analytics across all vendors and time periods
Admin Users	Super-admin only: create and manage admin accounts

5. Technical Architecture

NexConnect is built on a modern decoupled architecture separating the frontend presentation layer from the backend business logic via a RESTful API. This design enables independent scaling, technology evolution, and multi-platform delivery.

5.1 System Architecture Overview

Layer	Technology	Responsibility
Client Tier	Angular 19 (TypeScript)	4 standalone SPAs served separately. All state managed via Angular Signals.
API Gateway	Django REST Framework 3.17	Unified HTTP API at /api/* with JWT auth, pagination, filtering, CORS
Business Logic	Django 4.2 (Python)	Domain logic across 8 apps: accounts, products, orders, vendors, delivery, no
Authentication	SimpleJWT 5.5	Access tokens (1 day), refresh tokens (7 days with rotation), role-based perm
Background Jobs	Redis + RQ 2.10 + RQ Scheduler	Invoice generation, email delivery, task scheduling, order assignment
Push Notifications	Firebase Admin 6.5	Device token management, push notifications for order/delivery events
PDF Engine	ReportLab 4.0 + xhtml2pdf	Invoice generation: customer receipts, vendor settlements, delivery payouts
Database	PostgreSQL (psycopg2)	Primary relational data store. SQLite for local development.
File Storage	Django FileField / ImageField	Product images, vendor logos/banners, delivery photos, KYC documents
QR Codes	qrcode 8.0	Payment QR generation, OTP QR for delivery confirmation

5.2 Authentication & Authorisation

The platform implements a layered security model combining JWT token authentication with role-based access control (RBAC):

Role	Access Level	Key Permissions
customer	Customer App + API	Browse vendors/products, manage cart, place orders, raise issues
vendor	Vendor App + API	Manage own products/orders/coupons/profile (approved status required)
delivery	Delivery App + API	View/accept available orders, update delivery status, manage earnings
admin	Admin Panel + All APIs	Full platform access, user management, financial operations, config

Custom Permission Classes

IsAdminRole: user.role == "admin" — used on all /api/admin/* endpoints

IsVendor: user has a vendor_profile (any status) — for vendor profile read/update

IsApprovedVendor: vendor_profile.status == "approved" — for dashboard, products, orders

approvedVendorGuard (frontend): Checks localStorage.vendor_status; falls back to API call; redirects pending vendors

superuserGuard: Checks user.is_superuser — admin user management only

5.3 Multi-App Frontend Architecture

The frontend is structured as an Angular 19 monorepo using the Angular CLI workspace. All four applications share a single `node\_modules` and build pipeline while maintaining complete isolation for deployment.

Application	Port	Primary Users	Theme	Files
customer-app	4200	End customers	Light, Purple #6C63FF	71
vendor-app	4201	Store owners	Light, Purple #6C63FF	56
delivery-app	4202	Delivery partners	Light, Green #00C853	38
admin-panel	4203	Platform admins	Dark, Red #E94560 / Cyan #06BBD4	56
shared (lib)	N/A	All apps	Design tokens	14

Shared Library (@shared/public-api)

A compiled Angular library shared across all four apps providing: ApiService (all HTTP methods + reactive state), AuthService (login/register/logout with Signals), authInterceptor (automatic Bearer token injection), all route guards, and TypeScript model interfaces for every data entity.

5.4 Order Processing Workflow

Step	Action	System Behaviour
1	Cart Assembly	Customer adds items; cross-store guard prevents mixed-vendor carts with replace-cart dialog
2	Coupon Application	Real-time coupon validation at checkout; validates expiry, usage limits, min order amount
3	Order Placement	Multi-vendor cart splits into per-vendor sub-orders; coupon discount allocated proportionally; delivery fee calculated
4	Vendor Notification	Push notification + in-app notification sent to vendor on new order
5	Order Confirmation	Vendor updates status to confirmed → preparing → ready
6	Delivery Assignment	Admin or system assigns delivery partner; radius-expanding search algorithm
7	Pickup	Partner uses pickup OTP to confirm collection from vendor
8	Delivery	Partner delivers; customer provides delivery OTP; photo upload optional
9	Invoice	Background job generates PDF receipt and emails customer
10	Rating	Customer rates order (1–5 stars); updates delivery partner average rating
11	Issue (if any)	Customer raises return/refund/damage/mismatch; admin resolves via chat; refund initiated if applicable
12	Payout	Platform schedules vendor settlement; net = gross sales – commission; delivery partner earning logged

6. Technology Stack

6.1 Backend Technologies

Package	Version	Purpose
Django	4.2.29	Web framework — URL routing, ORM, admin, middleware
djangorestframework	3.17.0	RESTful API framework — serializers, viewsets, pagination
djangorestframework-simplejwt	5.5.1	JWT token authentication with rotation and blacklisting
django-cors-headers	4.9.0	Cross-origin resource sharing for frontend apps
django-filter	25.1	Advanced query filtering (DjangoFilterBackend)
psycopg2-binary	2.9.9	PostgreSQL database adapter
Pillow	12.1.1	Image processing — product images, avatars, delivery photos
reportlab	4.0.9	PDF generation — invoices, reports
xhtml2pdf	0.2.15	HTML template to PDF conversion
firebase-admin	6.5.0	FCM push notifications, device token management
django-rq	2.10.3	Redis Queue integration for background job processing
redis	5.0.3	In-memory data store — job queues, caching
rq-scheduler	0.13.1	Cron-like scheduled background tasks
qrcode	8.0	QR code generation for payment and OTP flows
PyJWT	2.12.1	JWT encoding/decoding utilities
cryptography	46.0.5	Encryption for sensitive fields (bank account numbers)

6.2 Frontend Technologies

Technology	Version	Purpose
Angular	19.0.0	Frontend framework — component architecture, routing, forms
TypeScript	5.6.2	Type-safe JavaScript for all frontend code
Angular Signals	19.0.0	Reactive state management replacing NgRx/BehaviorSubjects
RxJS	7.8.0	Reactive programming — HTTP streams, timers, subscriptions
Angular Router	19.0.0	Client-side routing with lazy loading and route guards
Angular Forms	19.0.0	Two-way data binding (FormsModule) for all forms

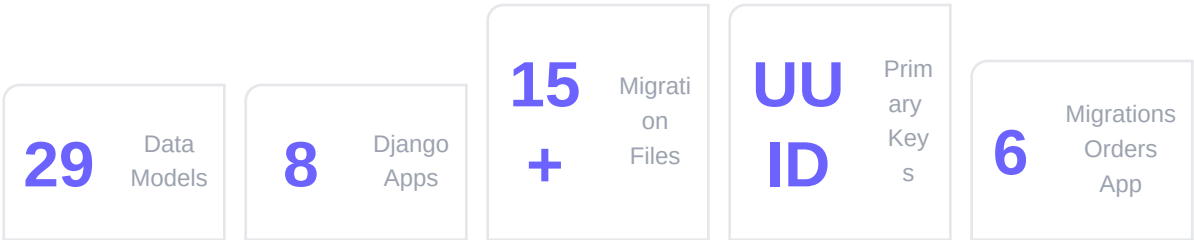
Technology	Version	Purpose
Angular HttpClient	19.0.0	HTTP communication with auto-interceptor token injection
Material Icons Outlined	Latest	Icon system — over 200 icons used across the platform
CSS Custom Properties	CSS3	Design token system for consistent theming across apps

6.3 Infrastructure & DevOps

Component	Technology	Notes
Web Server	Django Development Server	WSGI base (ASGI upgrade path available)
Database	PostgreSQL 15+	Primary production database; SQLite for local development
Cache / Queue	Redis 7.x	Background job broker and result store
Job Processing	RQ (Redis Queue)	Async: invoice generation, email dispatch, order assignment
Push Notifications	Firebase Cloud Messaging	Cross-platform push notifications
File Storage	Local filesystem	Upgrade path: AWS S3 / GCS via django-storages
Build System	Angular CLI 19	Monorepo workspace with per-app production builds
Containerisation	Docker-ready	docker-compose.yml for full local environment
Version Control	Git	Feature-branch workflow; CLAUDE.md documents conventions

7. Database Design & Data Models

The NexConnect database consists of 29 production data models across 8 Django applications. All primary keys are UUIDs for security and distributed-system compatibility. The schema supports full order lifecycle, vendor KYC, financial settlements, and dispute resolution.



Accounts App

Model	Key Fields
User	UUID PK, username, email, role (customer/vendor/delivery/admin), phone, avatar, is_verified, force_password_change, country
Address	UUID PK, user FK, label (home/work/other), full_name, phone, address lines, city, state, postal_code, lat/lng, is_default

Products App

Model	Key Fields
Category	UUID PK, name, slug, description, image, parent (self-FK for subcategories), is_active, display_order
Product	UUID PK, vendor FK, category FK, name, slug, description, price, compare_price, SKU, stock, unit, weight, is_available, is_featured, average_rating, total_ratings, total_orders
ProductImage	UUID PK, product FK, image, is_primary, is_ai_generated, display_order
ProductReview	UUID PK, product FK, customer FK, rating (1–5), comment — auto-updates product.average_rating

Orders App

Model	Key Fields
Cart	UUID PK, user OneToOne, computed total_items, total_amount
CartItem	UUID PK, cart FK, product FK, quantity, subtotal — unique_together (cart, product)
Order	UUID PK, order_number (ORD-XXXXXXXX), customer/vendor/partner FKs, status (8 states), pricing fields, coupon FK, OTP fields, delivery coords, placed_at

Model	Key Fields
OrderItem	UUID PK, order FK, product FK, immutable snapshots (product_name, product_price), quantity, subtotal
Coupon	UUID PK, code, discount_type (% , fixed, free_delivery), vendor FK (null=platform), usage limits, validity window
CouponUsage	UUID PK, coupon/user/order FKs, discount_applied, used_at
OrderRating	UUID PK, order OneToOne, customer/partner FKs, rating (1–5)
OrderTracking	UUID PK, order FK, status, description, lat/lng, timestamp
OrderIssue	UUID PK, order/customer FKs, issue_type (return/refund/damage/mismatch), status (5 states), refund_amount, refund_method, admin_notes, resolved_by FK
IssueMessage	UUID PK, issue FK, sender FK, is_admin flag, message, created_at

Vendors App

Model	Key Fields
Vendor	UUID PK, user OneToOne, store_name, description, logo/banner, contact, address, geolocation, tier, status, ratings, hours, min_order, delivery_radius
VendorOnboarding	UUID PK, vendor OneToOne, legal_name, tax/license fields, KYC docs, onboarding_status workflow
VendorBankDetails	UUID PK, vendor OneToOne, encrypted account_number, IFSC, bank_name, UPI, settlement_cycle, commission_percentage
VendorDocument	UUID PK, vendor FK, document_type (8 types), file, verification status, rejection_reason
VendorServiceableArea	UUID PK, vendor FK, pincode, city/state, is_active
VendorHoliday	UUID PK, vendor FK, date, reason
VendorAuditLog	UUID PK, vendor FK, action (15 types), description, performed_by FK, ip_address, metadata JSONField
VendorReview	UUID PK, vendor/customer FKs, rating (1–5), comment
VendorPayout	UUID PK, vendor FK, period, gross_sales, commission, net_payout, status (6 states), transaction_ref
DeliveryPartnerPayout	UUID PK, delivery_partner FK, period, total_deliveries, total_earnings, payout status

Delivery App

Model	Key Fields
DeliveryPartner	UUID PK, user OneToOne, vehicle_type/number, license, id_proof, is_approved, status (offline/available/on_delivery), current_lat/lng, ratings, earnings

Model	Key Fields
DeliveryAssignment	UUID PK, order OneToOne, status (searching/notified/accepted/etc), current_radius_km, notified/rejected/accepted partner M2M
DeliveryEarning	UUID PK, delivery_partner FK, order FK, amount
Asset	UUID PK, name, asset_type (5 types), serial_number, status (4 states), assigned_to FK

Notifications App

Model	Key Fields
Notification	UUID PK, user FK, title, message, notification_type (order/delivery/payout/promo/system), is_read, data JSONField
DeviceToken	UUID PK, user FK, FCM token (unique), platform

Support App

Model	Key Fields
SupportTicket	UUID PK, vendor FK, subject, category (5 types), priority (3 levels), message, status (4 states), admin_response, responded_by FK

Invoices App

Model	Key Fields
Invoice	UUID PK, invoice_number (auto: TYPE-XXXXXXX), invoice_type (3 types), order/vendor/recipient FKs, pdf_file, amount, tax_amount

8. API Architecture & Endpoints

All API communication occurs over REST/HTTP(S) at the base URL `http://host/api/`. The API uses JWT Bearer tokens for authentication, `PageNumberPagination` (20 items/page) globally, and supports `DjangoFilterBackend`, `SearchFilter`, and `OrderingFilter` on all list views.

Authentication — `/api/auth/`

Method	Path	Auth	Description
POST	<code>/api/auth/register/</code>	Public	Register new user (customer or vendor)
POST	<code>/api/auth/login/</code>	Public	Obtain JWT access + refresh tokens
GET PATCH	<code>/api/auth/profile/</code>	Authenticated	View or update own profile
POST	<code>/api/auth/change-password/</code>	Authenticated	Change password (with current password verification)
POST	<code>/api/auth/token/refresh/</code>	Public	Refresh expired access token
GET POST	<code>/api/auth/setup/</code>	Public (once)	Initial superuser creation

Vendors — `/api/vendors/`

Method	Path	Auth	Description
GET	<code>/api/vendors/</code>	Public	List all approved vendors (filterable, searchable)
GET	<code>/api/vendors//</code>	Public	Vendor detail with products list
GET	<code>/api/vendors/nearby/</code>	Public	Vendors within radius (lat/lng/radius_km params)
POST	<code>/api/vendors/register/</code>	Public	Vendor self-registration (returns tokens + vendor_status)
GET PATCH	<code>/api/vendors/profile/</code>	IsVendor	View/update own vendor profile
GET	<code>/api/vendors/dashboard/</code>	IsApprovedVendor	Revenue, order stats, top products
GET POST	<code>/api/vendors/products/</code>	IsApprovedVendor	Vendor product listing and creation
GET PUT PATCH DELETE	<code>/api/vendors/products//</code>	IsApprovedVendor	Product CRUD
GET	<code>/api/vendors/orders/</code>	IsApprovedVendor	Vendor order listing with status filter
PATCH	<code>/api/vendors/orders//status/</code>	IsApprovedVendor	Update order status

Method	Path	Auth	Description
GET POST PATCH DELETE	/api/vendors/coupons/	IsApprovedVendor	Vendor coupon CRUD

Products — /api/products/

Method	Path	Auth	Description
GET	/api/products/list/	Public	Product listing (vendor, category, search, price filters)
GET	/api/products//	Public	Product detail with reviews
GET	/api/products/featured/	Public	Featured/promoted products

Orders — /api/orders/

Method	Path	Auth	Description
GET	/api/orders/cart/	Authenticated	View current cart with items and totals
POST	/api/orders/cart/add/	Authenticated	Add product to cart
PATCH DELETE	/api/orders/cart/items//	Authenticated	Update quantity or remove item
DELETE	/api/orders/cart/clear/	Authenticated	Clear entire cart
POST	/api/orders/create/	Authenticated	Place order (splits by vendor, applies coupon, calculates delivery fee)
GET	/api/orders/list/	Authenticated	Customer order history with status filter
GET	/api/orders//	Authenticated	Order detail with items, tracking, partner info
POST	/api/orders//cancel/	Authenticated	Cancel order (only placed/confirmed status)
GET	/api/orders//tracking/	Authenticated	Order tracking events log
POST	/api/orders//rate/	Authenticated	Submit 1–5 star rating for delivered order
GET	/api/orders/coupons/	Authenticated	List active valid coupons for customer
POST	/api/orders/coupons/validate/	Authenticated	Validate coupon code with cart total
GET	/api/orders/issues/	Authenticated	List customer's own issues
POST	/api/orders/issues/	Authenticated	Raise new issue (return/refund/damage/mismatch)
GET	/api/orders/issues//	Authenticated	Issue detail with messages
POST	/api/orders/issues//messages/	Authenticated	Send chat message on issue

Admin — /api/admin/

Method	Path	Auth	Description
GET	/api/admin/stats/	IsAdminRole	Platform-wide user and revenue statistics
GET POST PATCH DELETE	/api/admin/vendors/	IsAdminRole	Full vendor management CRUD
GET PATCH	/api/admin/vendors//statuses/	IsAdminRole	Approve/reject/suspend vendor
PATCH	/api/admin/vendors//kyc-review/	IsAdminRole	Review KYC submission
GET PATCH	/api/admin/vendors//bank/	IsAdminRole	View/update bank details
GET POST	/api/admin/customers/	IsAdminRole	Customer management listing and detail
GET POST PATCH	/api/admin/delivery-partners/	IsAdminRole	Delivery partner onboarding and management
GET	/api/admin/orders/	IsAdminRole	All-platform orders with filters (status, vendor, customer, search)
GET PATCH	/api/admin/orders//	IsAdminRole	Order detail and status update
GET	/api/admin/issues/	IsAdminRole	All order issues with type/status/search filters
GET PATCH	/api/admin/issues//	IsAdminRole	Issue detail, status update, refund initiation
POST	/api/admin/issues//messages/	IsAdminRole	Admin reply in issue chat
GET POST PATCH DELETE	/api/admin/coupons/	IsAdminRole	Platform-wide coupon management
GET POST PATCH	/api/admin/payouts/vendors/	IsAdminRole	Vendor payout management
GET POST PATCH	/api/admin/payouts/delivery/	IsAdminRole	Delivery partner payout management
GET POST PATCH DELETE	/api/admin/categories/	IsAdminRole	Product category management
GET POST PATCH DELETE	/api/admin/products/	IsAdminRole	Platform-wide product management
GET POST PATCH DELETE	/api/admin/assets/	IsAdminRole	Delivery asset management
GET POST	/api/admin/notifications/	IsAdminRole	Send and manage notifications

9. Security & Compliance

9.1 Authentication Security

- JWT access tokens expire in 24 hours; refresh tokens expire in 7 days with automatic rotation
- Token blacklisting on logout prevents replay attacks
- Password change endpoint requires current password verification before update
- Force-password-change flag compels password reset for admin-created accounts (e.g. delivery partners)
- All API endpoints behind IsAuthenticated permission class by default
- Role-based permission classes prevent cross-role data access at the view layer

9.2 Data Protection

- Bank account numbers stored using field-level encryption (cryptography library)
- KYC documents stored in server filesystem with access-controlled paths
- Vendor audit log captures every sensitive operation with IP address and actor identity
- CORS headers configured (django-cors-headers) to restrict cross-origin access in production
- All UUID primary keys prevent enumeration attacks (non-sequential, unguessable)
- OTP verification for both pickup and delivery stages of order fulfilment

9.3 API Security

- authInterceptor automatically injects Bearer token on every frontend request
- CSRF protection via Django CsrfViewMiddleware
- Request validation via DRF serializers with field-level validators
- SQL injection prevention via Django ORM (parameterised queries)
- Rate limiting architecture: Redis-backed (upgrade path via django-ratelimit)
- Custom OptionalJWTAuthentication allows anonymous browsing of public endpoints without errors

9.4 Compliance Readiness

NexConnect is architected with Nigerian NDPR (Nigeria Data Protection Regulation) and CBN payment system guidelines in mind:

Requirement	Implementation Status
User data consent	Registration flow includes terms acceptance
KYC/AML for vendors	Full document verification workflow with audit trail

Requirement	Implementation Status
Financial record keeping	Immutable order records, payout history, invoice generation
Data access controls	Role-based access; audit logs on all sensitive admin actions
Dispute resolution	Formal return/refund/damage/mismatch workflow with documented resolution
Secure data transmission	HTTPS in production; all auth over encrypted transport

10. Business Model & Revenue Streams

NexConnect operates a marketplace model generating revenue from multiple complementary streams, reducing dependency on any single income source and creating a resilient financial foundation.

10.1 Revenue Streams

Revenue Stream	Mechanism	Rate / Model	Stage
Platform Commission	Percentage of vendor gross sales delivered to platform (configurable per vendor)	10-20% (configurable per vendor)	Anchor
Delivery Fee Revenue	Platform captures delivery fee margin	Base + 15% margin charge	Anchor
Vendor Subscription	Tiered monthly SaaS plan (basic/silver/gold/platinum)	₦5,000 - ₦50,000/mo features	Anchor
Coupon / Promotion Fees	Vendors pay for promoted placement on platform	Platform provides coupon	Blended
Transaction Processing	Micro-fee on each successfully completed transaction	0.5-1% per transaction	Roadmap
Data & Analytics	Anonymised market insights sold to B2B clients and distributors	₦2M/yr subscription	Roadmap
Delivery Insurance	Optional per-order insurance premium	1-2% of order value	Roadmap

10.2 Unit Economics

The following projections are based on conservative market penetration assumptions for the Lagos metropolitan area in Year 1:

Metric	Conservative	Base Case	Optimistic
Active Vendors (Y1)	500	1,000	2,500
Active Customers (Y1)	10,000	25,000	75,000
Orders/Month (Y1 run-rate)	15,000	40,000	120,000
Avg Order Value	₦3,500	₦4,200	₦5,500
GMV Monthly (Y1)	₦52.5M	₦168M	₦660M
Platform Revenue (15% take)	₦7.9M	₦25.2M	₦99M
Delivery Revenue	₦3.0M	₦8.0M	₦24M
Combined Monthly Revenue	₦10.9M	₦33.2M	₦123M

11. Operational Workflow

11.1 Vendor Onboarding

1. **Self-Registration:** Vendor registers on vendor app. Immediate login with status = "pending".
2. **KYC Submission:** Vendor submits legal name, business type, GST/PAN, FSSAI license, bank details, and KYC documents (PAN, GSTIN, cancelled cheque, identity proof, business registration).
3. **Document Review:** Admin reviews uploaded documents via admin panel. Approve/reject each document individually. Full audit trail logged.
4. **Bank Verification:** Admin verifies bank account details against submitted cancelled cheque.
5. **Final Approval:** Admin sets vendor status to "approved". Push notification sent to vendor. Vendor app auto-navigates to dashboard.
6. **Go Live:** Vendor can now create products, set prices/stock, configure delivery settings, and receive orders.

11.2 Delivery Partner Onboarding

Delivery partners cannot self-register (disabled endpoint returns 403). Admin creates partner accounts via the admin panel, uploads ID proof, assigns assets, sets vehicle details, and activates the account. Partner receives temporary password via the force_password_change mechanism.

11.3 Dispute Resolution

Phase	Actor	Action
1. Raise Issue	Customer	Selects issue type (return/refund/damage/mismatch), writes description. Issue created with status = "open".
2. Auto-Review	System	First customer message moves issue to "in_review" automatically.
3. Admin Review	Admin	Reviews issue in admin panel. Sends chat messages to customer for clarification.
4. Resolution	Admin	Sets status to: resolved / rejected / refund_initiated. If refund: enters amount and method.
5. Customer Notification	System	Status update visible in customer's My Issues screen. Refund chip displayed if initiated.
6. Payout Adjustment	Admin	Refund handled manually via bank transfer or wallet credit (payment gateway integration on roadmap).

12. Scalability & Infrastructure

NexConnect is architected for horizontal scale from day one. The decoupled frontend/backend model, stateless JWT authentication, and Redis-backed job queuing form the foundation for a cloud-native deployment.

Scaling Dimension	Current State	Scale Path
Application Servers	Single Gunicorn instance	Load-balanced behind Nginx; auto-scaling group on AWS/GCP
Database	PostgreSQL single instance	Read replicas for analytics; connection pooling via PgBouncer
Job Processing	Single RQ worker	Multiple RQ workers; RQ Scheduler for recurring tasks
File Storage	Local filesystem	AWS S3 / Google Cloud Storage via django-storages (1-day migration)
CDN	None (dev)	CloudFront or Cloudflare for static assets and media
Cache Layer	Redis (jobs only)	Add Django cache framework for API response caching
Search	Django ORM + filters	Elasticsearch upgrade path for full-text product search
Notifications	Firebase FCM + DB	Websocket upgrade for real-time chat (Django Channels)
Frontend Hosting	Dev server	Static build on AWS S3 + CloudFront (PWA-ready)

Performance Design Decisions

- Pagination enforced globally at 20 items per page — prevents large payload responses
- `select_related` and `prefetch_related` used in all complex queriesets — eliminates N+1 database queries
- Background job processing (invoice generation, email) — decouples order placement latency from async work
- Lazy-loaded Angular routes — each app page loaded on demand, reducing initial bundle size
- Angular Signals — fine-grained reactivity without full component re-renders
- Polled real-time updates (5s customer, 15s admin) — low infrastructure overhead versus WebSockets
- UUID primary keys — distributed-safe, no centralised sequence bottleneck

13. Development Roadmap

NexConnect follows a phased development strategy, ensuring core platform stability before adding growth-oriented features.

Phase	Timeline	Deliverables	Status
Phase 0 Foundation	Completed	Django backend, 4 Angular apps, auth, vendor/customer flows, order lifecycle, basic admin	✓ Done
Phase 1 Commerce	Completed	Product catalogue, cart, checkout, delivery fee engine, order tracking, invoice PDF, vendor ratings	✓ Done
Phase 2 Growth	Completed	Coupon & discount engine (% , fixed, free delivery), vendor-specific and platform-wide coupons, offers page	✓ Done
Phase 3 Trust & Safety	Completed	Order issue system (return/refund/damage/mismatch), live chat, refund initiation, My Issues customer hub	✓ Done
Phase 4 Payments	Q3 2025	Paystack/Flutterwave integration, online payment, automated refunds, wallet system	■ Planned
Phase 5 Retention	Q4 2025	Loyalty points, referral rewards, personalised recommendations, repeat order shortcuts	■ Planned
Phase 6 Expansion	Q1 2026	Multi-city support, vendor franchising, B2B bulk ordering, API marketplace for 3rd-party integrations	■ Planned
Phase 7 Enterprise	Q2 2026	White-label platform offering, enterprise vendor contracts, data analytics dashboard / Metabase	■ Planned

14. Team & Governance

NexConnect is built by a focused founding team combining deep technical expertise with on-the-ground market knowledge. The current MVP represents thousands of engineering hours delivering a production-grade platform.

Technical Capabilities Demonstrated

Full-Stack Engineering: Django REST Framework backend + Angular 19 frontend — end-to-end ownership of the entire technical stack

Systems Architecture: 8-app Django monolith with clear domain boundaries, shared library pattern, and role-based security model

Product Design: Four distinct UX paradigms (mobile customer, desktop vendor, mobile delivery partner, dark admin) in a unified design system

DevOps Readiness: Docker-ready infrastructure, PostgreSQL, Redis, background job processing, and cloud storage upgrade paths

Security Engineering: JWT auth, field-level encryption, role-based access control, audit logging, OTP verification

Financial Systems: Commission calculation, payout workflows, coupon discount engine, invoice PDF generation

Governance Structure

The platform is structured for transparent governance with full audit trails. Every vendor status change, document verification, bank detail update, and admin action is recorded in the VendorAuditLog with timestamp, actor identity, and IP address. Superuser accounts have additional protection via a dedicated guard, separating administrative operations from platform governance.

15. Financial Projections

The following 3-year financial projections are based on a phased market entry strategy starting in Lagos, expanding to Abuja and Port Harcourt in Year 2, and pan-Nigeria coverage in Year 3. All figures in Nigerian Naira (₦).

15.1 Growth Assumptions

Parameter	Year 1	Year 2	Year 3
Active Vendors	1,000	5,000	20,000
Active Customers	25,000	150,000	600,000
Monthly Orders	40,000	250,000	1,200,000
Avg Order Value	₦4,200	₦5,000	₦6,000
Monthly GMV	₦168,000,000	₦1,250,000,000	₦7,200,000,000
Cities Covered	1 (Lagos)	3	10+
Delivery Partners	500	3,000	15,000

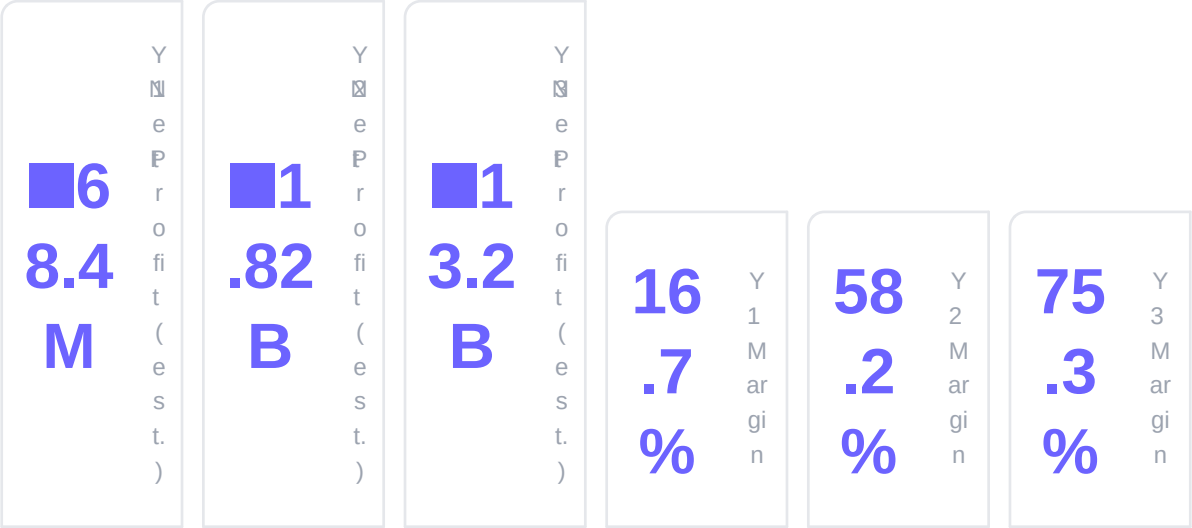
15.2 Revenue Projections

Revenue Source	Year 1 (₦)	Year 2 (₦)	Year 3 (₦)
Platform Commission (15%)	302,400,000	2,250,000,000	12,960,000,000
Delivery Fee Revenue	96,000,000	600,000,000	2,880,000,000
Vendor Subscriptions	—	150,000,000	960,000,000
Coupon/Promotion Fees	12,000,000	125,000,000	720,000,000
TOTAL REVENUE	₦410,400,000	₦3,125,000,000	₦17,520,000,000

15.3 Cost Structure

Cost Category	Year 1 (₦)	Year 2 (₦)	Year 3 (₦)
Engineering & Product	180,000,000	480,000,000	960,000,000
Cloud Infrastructure	24,000,000	120,000,000	480,000,000
Operations & Support	36,000,000	180,000,000	720,000,000
Sales & Marketing	60,000,000	375,000,000	1,752,000,000

Cost Category	Year 1 (■)	Year 2 (■)	Year 3 (■)
Legal & Compliance	18,000,000	60,000,000	144,000,000
General & Admin	24,000,000	90,000,000	264,000,000
TOTAL COSTS	■342,000,000	■1,305,000,000	■4,320,000,000



16. Risk Analysis

Risk	Category	Likelihood	Impact	Mitigation
Payment gateway delays	Technical	Medium	High	COD-first model live now; Paystack integration in Q3 roadmap
Low vendor adoption	Commercial	Medium	High	Free onboarding, zero-commission first 3 months, dedicated onboard
Delivery partner retention	Operational	Medium	Medium	Earnings transparency, flexible scheduling, asset provision
Regulatory changes (CBN/Regulatory)	Regulatory	Low	High	Legal counsel retained; KYC/AML-compliant architecture
Cybersecurity breach	Technical	Low	Critical	JWT auth, field encryption, audit logs, penetration testing planned
Competitor entry (Jumia, Glovo)	Competitive	High	Medium	Hyperlocal focus, vendor-specific features, faster onboarding
Infrastructure downtime	Technical	Low	High	Cloud redundancy, database backups, load balancing on roadmap
FX/Naira volatility	Financial	High	Low	All-NGN operations; FX hedging not required until cross-border expansion

17. Conclusion & Investment Ask

NexConnect is not a concept — it is a fully functional, production-ready platform. The MVP has been built with institutional-grade engineering practices, comprehensive feature coverage, and a clear path to monetisation and scale.

What Has Been Built

- 4 separate, deployable web applications across 3 user roles + admin
- 29 data models with full relational integrity, UUID primary keys, and migration history
- 60+ REST API endpoints with role-based access control and JWT authentication
- Complete order lifecycle: discovery → cart → checkout → delivery → rating → dispute resolution
- Vendor KYC onboarding with document verification, audit trail, and payout engine
- Coupon/discount engine supporting percentage, fixed-amount, and free-delivery promotions
- Dispute resolution system: raise issues, live admin chat, refund initiation
- Delivery partner management: assignment algorithm, OTP verification, earnings tracking
- Background job processing for invoice PDF generation and email delivery
- Firebase push notification integration and in-app notification system

Investment Ask

We are seeking an initial investment to fuel the next phase of growth — payment gateway integration, marketing for vendor and customer acquisition, and infrastructure scale-out for production launch in Lagos.

Use of Funds	Allocation	Amount (₦)
Payment Gateway Integration (Paystack/Flutterwave)	15%	15,000,000
Cloud Infrastructure & DevOps	20%	20,000,000
Vendor Acquisition & Onboarding Campaign	25%	25,000,000
Customer Acquisition (Digital Marketing)	20%	20,000,000
Engineering Team Expansion (2 engineers)	15%	15,000,000
Legal, Compliance & Regulatory	5%	5,000,000
TOTAL SEED ROUND	100%	₦100,000,000

NexConnect represents a rare combination of technical execution excellence and clear market opportunity. The platform is ready — the market is ready. We invite you

to be part of building Nigeria's most trusted local delivery ecosystem.

NexConnect Technologies
Lagos, Nigeria

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Prepared: April 06, 2026

MVP Status: **LIVE**
Platform: **Web (PWA-ready)**